1

$n=2 \qquad \psi_2(x) = \sqrt{\frac{2}{a}} \sin \frac{2\pi x}{a}$

$$P = \int_{\frac{2a}{3}}^{a} |\psi_2(x)|^2 dx = \frac{2}{a} \int_{\frac{2a}{3}}^{a} \sin^2\left(\frac{2\pi x}{a}\right) dx =$$

$$= \left(y = \frac{2\pi}{a} x\right) = \frac{2}{a} \frac{a}{2\pi} \int_{\frac{4\pi}{3}}^{2\pi} \sin^2(y)\, dy =$$

$$\sin^2(y) = \frac{1}{2}(1 - \cos 2y)$$

$$= \frac{1}{2\pi} \int_{4\pi/3}^{2\pi} (1 - \cos 2y)\, dy = \frac{1}{2\pi} \left\{ 2\pi - \frac{4\pi}{3} - \frac{1}{2} \sin 2y \Big|_{4\pi/3}^{2\pi} \right\}$$

$$= \frac{1}{2\pi} \left\{ \frac{2\pi}{3} - \frac{1}{2}\left(\sin 4\pi - \sin \frac{8\pi}{3}\right) \right\} = \frac{1}{3} + \frac{\sqrt{3}}{8\pi}.$$

$$\sin \frac{8\pi}{3} = \sin \frac{9\pi - \pi}{3} = \sin\left(3\pi - \frac{\pi}{3}\right) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

2 $\Psi(x,0) = A \sin^3 \frac{\pi x}{a} \qquad (0 \le x \le a)$

(a) Need to ~~find~~ express $\sin^3\theta$ via linear combinations.

$$\sin^3\theta = \sin\theta \cdot \sin^2\theta = \sin\theta \cdot \frac{1}{2}(1-\cos 2\theta) =$$

$$= \frac{1}{2}\left(\sin\theta - \sin\theta \cdot \cos 2\theta\right)$$

$$\sin\theta \cdot \cos 2\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)\frac{1}{2}\left(e^{i2\theta} + e^{-i2\theta}\right) =$$

$$= \frac{1}{4i}\left(e^{i3\theta} + e^{-i\theta} - e^{i\theta} - e^{-i3\theta}\right) =$$

$$= \frac{1}{4i}\left(2i \sin 3\theta - 2i \sin\theta\right) = \frac{1}{2}\left(\sin 3\theta - \sin\theta\right)$$

$$\Rightarrow \quad \sin^3\theta = \frac{1}{2}\left(\sin\theta - \frac{1}{2}\sin 3\theta + \frac{1}{2}\sin\theta\right) =$$

$$= \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta$$

We have $\theta = \frac{\pi x}{a} \Rightarrow$

$$\Psi(x,0) = \frac{A}{4}\left(3\sin\frac{\pi x}{a} - \sin\frac{3\pi x}{a}\right)$$

but $\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right) \Rightarrow \sin\left(\frac{n\pi x}{a}\right) = \sqrt{\frac{a}{2}}\,\psi_n(x)$

$$\Rightarrow \Psi(x,0) = \frac{A}{4}\sqrt{\frac{a}{2}}\left[3\psi_1(x) - \psi_3(x)\right].$$

(b) Normalize using orthogonality of different ψ_n's:

$$1 = \left(\frac{A}{4}\sqrt{\frac{a}{2}}\right)^2 \int_0^a |3\psi_1 - \psi_3|^2\,dx = \frac{A^2 a}{32}\left\{9\int|\psi_1|^2dx\right.$$

$$\left. + \int|\psi_3|^2dx\right\} = \frac{A^2 a}{32}\cdot(9+1) = \frac{5a}{16}A^2 \Rightarrow A = \sqrt{\frac{16}{5a}} = \frac{4}{\sqrt{5a}}$$

So that $\Psi(x,0) = \frac{4}{\sqrt{5a}} \frac{1}{4}\sqrt{\frac{a}{2}} \left(3\psi_1 - \psi_3\right) =$

$= \frac{1}{\sqrt{10}} \left(3\psi_1 - \psi_3\right)$

(c) $\Psi(x,t) = \frac{3}{\sqrt{10}} \psi_1(x) e^{-iE_1 t/\hbar} - \frac{1}{\sqrt{10}} \psi_3(x) e^{-iE_3 t/\hbar}$

where $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \Rightarrow E_3 = 9E_1$.

3 $\Psi(x,0) = \frac{1+i}{2}\psi_0(x) + \frac{1}{\sqrt{2}}\psi_1(x)$

(a) $\Psi(x,t) = \frac{1+i}{2}\psi_0(x)\, e^{-iE_0 t/\hbar} + \frac{1}{\sqrt{2}}\psi_1(x)\, e^{-iE_1 t/\hbar}$

where $E_n = \hbar\omega\left(n+\frac{1}{2}\right)$ are energy levels of harmonic oscillator.

$\Rightarrow \Psi(x,t) = \frac{1+i}{2}\psi_0(x)\, e^{-i\omega t/2} + \frac{1}{\sqrt{2}}\psi_1(x)\, e^{-i3\omega t/2}$

(b) From general expression $\Psi(x,t) = \sum_{n=0}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}$ we know that probability to observe energy E_n (eigen value of the problem!) is given by $|c_n|^2$.

$\frac{1}{2}\hbar\omega = E_0 \Rightarrow$ probability to get it is

$$P_0 = |c_0|^2 = \left(\frac{1+i}{2}\right)^2 = \frac{1+1}{4} = \frac{1}{2}.$$

$\frac{3}{2}\hbar\omega = E_1 \Rightarrow$ probability $P_1 = |c_1|^2 = \frac{1}{2}$.

$\frac{7}{2}\hbar\omega = E_3 \Rightarrow$ probability $P_3 = |c_3|^2 = 0$ because only c_0 and c_1 are non-zero in this problem.

(c) $\langle x\rangle = \int_{-\infty}^{\infty} x\, |\Psi(x,t)|^2\, dx =$

$$= \int \Psi^*(x,t)\, x\, \Psi(x,t)\, dx = \int \left(\frac{1-i}{2}\psi_0^* e^{i\omega t/2} + \frac{1}{\sqrt{2}}\psi_1^* e^{i3\omega t/2}\right) \cdot x \left(\frac{1+i}{2}\psi_0 e^{-i\omega t/2} + \frac{1}{\sqrt{2}}\psi_1 e^{-i3\omega t/2}\right)$$

$$\langle x\rangle = \frac{(1-i)(1+i)}{4}\int \underbrace{\psi_0^* x\, \psi_0\, dx}_{0} + \frac{1}{\sqrt{2}\cdot\sqrt{2}}\int \underbrace{\psi_1^* x\, \psi_1\, dx}_{0}$$

$$+ \frac{1-i}{2\sqrt{2}}\int \psi_0^* x\, \psi_1\, e^{-i\omega t} dx + \frac{1+i}{2\sqrt{2}}\int \psi_1^* x\, \psi_0\, e^{i\omega t} dx .$$

Now $x = \sqrt{\frac{\hbar}{2m\omega}}\,(a_+ + a_-)$

and $(a_+ + a_-)\,\psi_1 = \sqrt{2}\,\psi_2 + \psi_0$

$(a_+ + a_-)\,\psi_0 = \psi_1$

but $\int \psi_0^* \psi_2\, dx = 0 \Rightarrow$

$$\langle x\rangle = \frac{1-i}{2\sqrt{2}}\sqrt{\frac{\hbar}{2m\omega}}\, e^{-i\omega t}\underbrace{\int \psi_0^*\psi_0\, dx}_{1} + \frac{1+i}{2\sqrt{2}}\sqrt{\frac{\hbar}{2m\omega}}\, e^{i\omega t}\underbrace{\int \psi_1^*\psi_1\, dx}_{1}$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{1-i}{2\sqrt{2}}\, e^{-i\omega t} + \frac{1+i}{2\sqrt{2}}\, e^{i\omega t}\right) =$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{2\cos\omega t}{2\sqrt{2}} + \frac{i}{2\sqrt{2}}\, 2i\sin\omega t\right) =$$

$$= \sqrt{\frac{\hbar}{4m\omega}}\left(\cos\omega t - \sin\omega t\right).$$

Similarly $\langle p\rangle = \frac{1-i}{2\sqrt{2}}\int \psi_0^*\, p\, \psi_1\, e^{-i\omega t} dx + \frac{1+i}{2\sqrt{2}}\int \psi_1^*\, p\, \psi_0\, e^{i\omega t} dx$

Using $p = i\sqrt{\frac{\hbar m\omega}{2}}\,(a_+ - a_-)$

and $(a_+ - a_-)\,\psi_1 = \sqrt{2}\,\psi_2 - \psi_0$

$(a_+ - a_-)\,\psi_0 = \psi_1$

$$\langle p\rangle = i\sqrt{\frac{\hbar m\omega}{2}}\left\{\frac{1-i}{2\sqrt{2}}\, e^{-i\omega t}\underbrace{\int \psi_0^*(-\psi_0)\,dx}_{-1} + \right.$$

$$\left. + \frac{1+i}{2\sqrt{2}}\, e^{i\omega t}\underbrace{\int \psi_1^*\psi_1\,dx}_{1}\right\} = i\sqrt{\frac{\hbar m\omega}{2}}\,\frac{1}{2\sqrt{2}}\left((1+i)\,e^{i\omega t} - (1-i)\,e^{-i\omega t}\right)$$

$$= \sqrt{\frac{\hbar m\omega}{2}}\,\frac{i}{2\sqrt{2}}\left(\underbrace{e^{i\omega t} - e^{-i\omega t}}_{2i\sin\omega t} + i\underbrace{(e^{i\omega t} + e^{-i\omega t})}_{2\cos\omega t}\right) =$$

$$= \sqrt{\frac{\hbar m\omega}{4}}\left(-\sin\omega t - \cos\omega t\right)$$

Note that $\langle p\rangle = m\dfrac{d\langle x\rangle}{dt}$.

(d) Ehrenfest's theorem states $\dfrac{d\langle p\rangle}{dt} = \left\langle -\dfrac{dV}{dx}\right\rangle$

Here $-\dfrac{dV}{dx} = -\dfrac{d}{dx}\left(\dfrac{1}{2}m\omega^2x^2\right) = -m\omega^2 x$.

$\Rightarrow$ theorem reads $\dfrac{d\langle p\rangle}{dt} = -m\omega^2\langle x\rangle$

Check it: $\dfrac{d}{dt}\langle p\rangle = \sqrt{\dfrac{\hbar m\omega}{4}}\,\dfrac{d}{dt}\left(-\sin\omega t - \cos\omega t\right) =$

$$= \sqrt{\frac{\hbar m\omega}{4}}\;\omega\left(\sin\omega t - \cos\omega t\right)$$

While $-m\omega^2\langle x\rangle = \left(-m\omega^2\right)\sqrt{\dfrac{\hbar}{4m\omega}}\left(\cos\omega t - \sin\omega t\right) =$

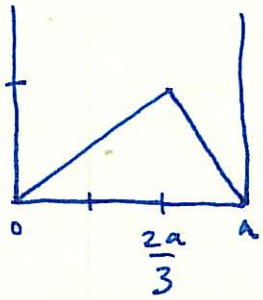
$$= \sqrt{\frac{\hbar m\omega}{4}}\,(\omega)\left(\sin\omega t - \cos\omega t\right)$$

Thus indeed $\dfrac{d}{dt}\langle p\rangle = -m\omega^2\langle x\rangle$ is satisfied.

4

$$\Psi(x,0) = \begin{cases} A\,x/2 & 0 \le x \le 2a/3 \\ A(a-x) & \frac{2a}{3} \le x \le a \end{cases}$$

(a)



Normalization $\quad 1 = \int_0^a |\Psi|^2 dx$

$$\Rightarrow \quad 1 = A^2 \left\{ \int_0^{2a/3} \left(\frac{x}{2}\right)^2 dx + \int_{\frac{2a}{3}}^{a} (a-x)^2 dx \right\} =$$

$$= A^2 \left\{ \frac{1}{4} \frac{x^3}{3} \Big|_0^{\frac{2a}{3}} - \int_{\frac{a}{3}}^{0} \underset{y=a-x}{y^2 dy} \right\} =$$

$$= A^2 \left\{ \frac{1}{12} \left(\frac{2a}{3}\right)^3 + \frac{1}{3} \left(\frac{a}{3}\right)^3 \right\} = A^2 a^3 \left(\frac{8}{4 \cdot 3^4} + \frac{1}{3^4} \right) = \frac{A^2 a^3}{27}$$

$$\Rightarrow \quad A = \sqrt{\frac{27}{a^3}} .$$

(b)

$$\Psi(x,0) = \sqrt{\frac{27}{a^3}} \begin{cases} \frac{x}{2} , & 0 \le x \le \frac{2a}{3} \\ a-x , & \frac{2a}{3} \le x \le a \end{cases}$$

$$\Rightarrow \frac{d\Psi}{dx} = \sqrt{\frac{27}{a^3}} \begin{cases} \frac{1}{2} , & 0 \le x \le \frac{2a}{3} \\ -1 , & \frac{2a}{3} \le x \le a \end{cases}$$

Let us fit this to the form: $\frac{d\Psi}{dx} = \left(\frac{27}{a^3}\right)^{1/2} \left(\alpha + \beta\, \theta\left(x - \frac{2a}{3}\right) \right)$

$$\theta\left(x - \frac{2a}{3}\right) = \begin{cases} 0 , & x < \frac{2a}{3} \\ 1 , & x > \frac{2a}{3} \end{cases}$$

$$\Rightarrow \quad \frac{d\Psi}{dx} = \left(\frac{27}{a^3}\right)^{1/2} \begin{cases} \alpha , & 0 \le x \le \frac{2a}{3} \\ \alpha + \beta , & \frac{2a}{3} \le x \le a \end{cases}$$

$$\Rightarrow \alpha = \frac{1}{2} , \quad \alpha + \beta = -1 \quad \Rightarrow \quad \beta = -\frac{3}{2} .$$

Thus $\frac{d\Psi}{dx} = \sqrt{\frac{27}{a^3}}\left(\frac{1}{2} - \frac{3}{2}\,\theta\left(x - \frac{2a}{3}\right)\right).$

(c) Using $\frac{d\,\theta(x-x_0)}{dx} = \delta(x-x_0)$ we obtain

$$\frac{d^2\Psi}{dx^2} = \sqrt{\frac{27}{a^3}}\left(-\frac{3}{2}\right)\delta\left(x - \frac{2a}{3}\right).$$

(d) $$\langle H\rangle = \int_0^a \Psi(x,0)^*\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right)\Psi(x,0)\,dx$$

$$= -\frac{\hbar^2}{2m}\int_0^a \Psi^*(x,0)\left(-\frac{3}{2}\sqrt{\frac{27}{a^3}}\right)\delta\left(x - \frac{2a}{3}\right)dx$$

$$= \frac{3\hbar^2}{4m}\sqrt{\frac{27}{a^3}}\;\Psi^*\left(\frac{2a}{3},0\right) = \frac{3\hbar^2}{4m}\sqrt{\frac{27}{a^3}}\sqrt{\frac{27}{a^3}}\;\frac{1}{2}\cdot\frac{2a}{3} =$$

$$= \frac{27\hbar^2}{4ma^2}.$$

(e) Consider arbitrary smooth function $f(x)$:

$$I = \int_{-\infty}^{+\infty}\frac{d\theta(x-x_0)}{dx}f(x)\,dx = \theta(x-x_0)f(x)\Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty}\theta(x-x_0)\frac{df}{dx}dx$$

but $\theta(x-x_0) = \begin{cases}1, & x > x_0\\ 0, & x < x_0\end{cases}$

$$\Rightarrow I = f(+\infty) - \int_{x_0}^{\infty}\frac{df}{dx}dx = f(+\infty) - \left[f(+\infty) - f(x_0)\right] = f(x_0)$$

$$= \int f(x)\,\delta(x-x_0)\,dx \Rightarrow \frac{d\theta(x-x_0)}{dx} = \delta(x-x_0)$$